

The Circle: 7 different perspectives

Roshan Raj

06 March 2024

A circle is a uniform and beautiful curved line without corners or cracks, all its points in the plane. This not being a formal definition. Still, this figure is a very part of our knowledge and civilization, as it appears in an outline of the sun and the human eyes and various depictions are found in petroglyphs and cave paintings, from Egyptian Rhind papyrus to the Dharma wheel and mandalas, in the books of Sulbhasutra and Euclid's Elements. This is a transcendental figure, simple yet profound.

In this article, we will encounter different ways to understand the circle. How the ordinary geometry and a relatively tricky algebraic topology perceive and define the exact figure uniquely. A series of lectures by NJ Wildberger on algebraic topology is recommended for interested readers.

As per geometry

A circle $\mathbb{S}^1$, in the Euclidean plane, is defined by a polynomial, $(x - a)^2 + (y - b)^2 - r^2 = 0$. The radius is r , and the centre is at $c = (a, b)$, expressing the amount by which the circle's centre c is shifted from the origin $(0, 0)$. In tracing out the outline of a circle, we can happily and arguably introduce the idea of a parametric approach. In an attempt to make a stereographic projection of the circle ((see figure 1), we have $e(h) = \left(\frac{1-h^2}{1+h^2}, \frac{2h}{1+h^2}\right)$. It is like opening the circle and projecting the complete circle over the real number line.

Now, this so-called *rational parametrization*, can be seen to provide its algebraic analogue: *tran-*

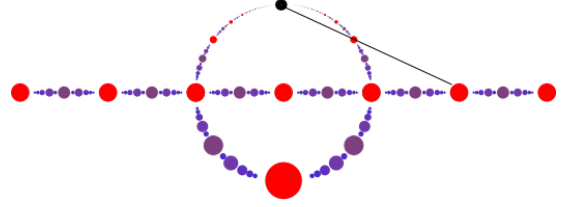


Figure 1: Stereographic projection of rational points on the circle (Wikimedia Commons)

scendental parametrization as $\rho(\theta) = (\cos \theta, \sin \theta)$: Your probable acquaintance may have occurred after studying wave propagation; where around a similar note, we open the circle on the sinusoidal wave rather on a line. Talking about the parameters, we see h is any element $\mathbb{R}$ along with ∞ , and θ is a closed parameter $0 \leq \theta \leq 2\pi$. With such rational parametrization, the one we describe does bear a point a $h = 0$, and its diametrically opposite point is calibrated to be $h = \infty$.

As per complex number

In complex plane, where $z = re^{i\theta} + c = r(\cos \theta + i \sin \theta) + c$, it can define a circular loop by $|z - c| = r$, Where r is radius and c is the centre. One can relate its link to geometry when on invoking the representation of a unit circle on the Argand plane.

As per algebraic topology

In algebraic topology, a circle $\mathbb{S}^1$ isn't defined geometrically with a centre and radius but rather through the concept of quotients in topological spaces (the

finest topology that makes the projection map to its equivalence class continuous):

- The quotient of Real Numbers by Integers: The circle is the result of taking all real numbers and identifying any two numbers that differ by an integer. Imagine a number line stretching infinitely in both directions. Glueing the ends together (identifying 1 with 0, 2 with 1, and so on) creates a loop - the circle.
- The quotient of the Unit Interval: Another way to view it is as the quotient of the closed unit interval $[0, 1]$ by the equivalence relation that identifies 0 and 1. Here, we take a line segment of length 1 and connect its endpoints, again forming a loop- the circle.

Both these approaches are mathematically equivalent and give us the circle as a topological space. This space inherits its topological properties (like continuity and connectedness) from the real numbers or the unit interval.

In short, a circle is typically described as a result of taking the quotient space of the unit circle's circumference, where points are considered equivalent based on a specific equivalence relation. This relation often encompasses rotations and reflections.

As per moduli space

Consider an affine plane $\mathbb{A}^2$; the space of all one-dimensional subspace $\mathbb{P}^1$, defines circle $\mathbb{S}^1$. The reason is simple: as a one-dimensional subspace consists of lines (projective lines $\in P^1$) passing through origins, all such sets of infinitely many lines would make out a circle. Conversely, we can bisect a given circle diametrically through infinitely many diameters. Then, each diameter is, in fact, a line through the origin, which again is an element of a one-dimensional subspace of $\mathbb{A}^2$. Through such prescription, the circle is not like infinity-thin pizza slices, but a circle

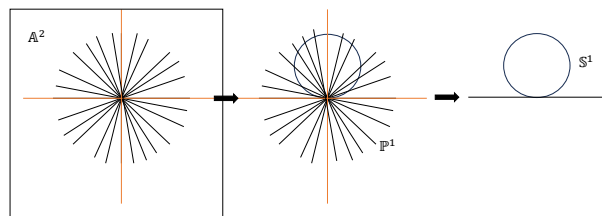


Figure 2: Description of a circle as per moduli space

with an x-axis as its only tangent, and the rest of the bisectors are projective lines. It may require you to look at the illustration 2.

I insist you recall the stereographic projection of the circle, which is told to be related as if the circle is opened. Merge this idea with the above description of a circle with an x-axis as its only tangent. Notice that both are related. If we are to mark the tangent point as $h = 0$, then its diametrically opposite point would be projected at infinity at ∞ and $-\infty$ on the x-axis. Doesn't it sound like rational parametrization? Yes, it does.

As per polygonal representation

Topologically, a circle $\mathbb{S}^1$ is any closed polygon or, formally, the topological space that is homeomorphic (topological isomorphism) to the unit circle in the Euclidean plane ($\mathbb{R}^2$). In short, any polygons can be deformed (topologically) into a circle in finite steps. Hence, all polygons are topologically invariant to the circle.

One must ask, is a line and a circle find common ground? It is known that a line is infinitely thin, and its ends extend to infinity, but the circle is compact. One should learn that compactness is one of the topological properties. Technically, There does exist a continuous bijection between affine line $\mathbb{A}^1$

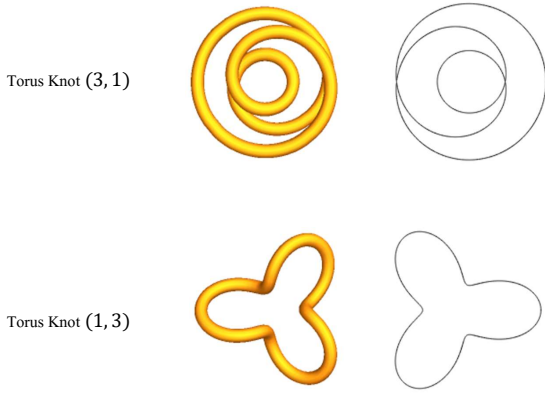


Figure 3: Description of a circle as per Knot Theory and circle $\mathbb{S}^1$, but its inverse has a discontinuity at 0. Hence, a line is not topologically invariant to the circle due to compactness.

As per knot theory

A circle is an unknot in $\mathbb{A}^3$ space! A more sophisticated definition can be obtained. Let p and q be two relatively prime, then a torus knot with parameters (p, q) is formed by threading a string (rope) through a torus's opening p times while making q complete revolutions before connecting the ends. As a result of this definition, a circle is a particular case of a torus knot with no internal nodes or all $(1, q) = (p, 1)$ -torus knots, as per equivalence properties.

As per translation on a line

One should accept (without proof) that a circle carries a group structure, which is abelian and non-cyclic, $S^1 = \{z \in \mathbb{C} : |z| = 1\}$, is called a *circle group*. If defining *unit translation on a affine line* $\mathbb{A}^1$ towards right as τ and its inverse τ^{-1} (towards left). We can finally perceive circle as $S^1 \approx \frac{\mathbb{A}^1}{\langle \tau, \tau^{-1} \rangle}$.

To get a pictorial understanding based on the above expression, one can introduce the concept of *orbit* consisting of those unit translations (τ and τ^{-1}) and all its iterates (multiple) on $\mathbb{A}^1$. We define a

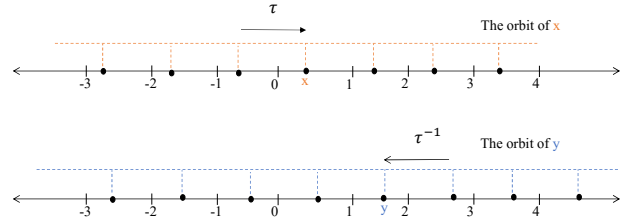


Figure 4: Orbit of points x and y on affine line $\mathbb{A}^1$.

family of the orbit of point x (red) and the orbit of y (blue); see figure (4). Suppose, when in due course, varying x passes through y , then moving (translating) orbits will get us back to the initial situation at the next point $\tau(x)$. We note all such orbits between x and $\tau(x)$ are distinct, but the ends are the same and produce the same properties. Thus, we induce that the *space of all orbits to be a family of circle*. More intuitively, an n -iteration within the closed interval is similar to tracing over the circle n -times, where the initial and final point corresponds to x and $\tau(x)$. Also, y is any other point on the circle. This correspondence between affine line and circle should not be taken as if $\mathbb{A}^1$ and $\mathbb{S}^1$ are homeomorphic, and in fact, they are not.

Conclusion

In this article, though non-rigorously, we have explored the concept of a circle from seven different perspectives, each shedding light on its unique characteristics and mathematical interpretations. From the geometric representation defining a circle in terms of its centre and radius to the algebraic topology's quotient space approach and from the complex plane representation to its connection with moduli space and knot theory, we have witnessed the circle's versatility and depth. Additionally, the polygonal representation's insight into topological invariance and the abstraction of the circle as a group in translation on a

line further enrich our understanding. Through these diverse viewpoints, we appreciate the circle as not merely a geometric shape but a fundamental mathematical concept with profound implications across various fields.

Roshan Raj
Department of Physics, Sardar Vallabhbhai National
Institute of Technology, Surat, Gujarat, India-395007
E-Mail Id: roshankaalkrit@gmail.com